

4.5 Additional Tools for Evaluating Binary Classifiers

For classification, and in particular binary classification, there are a wide range of additional tools for inspecting the performance beyond the misclassification rate. For simplicity we consider the binary problem and use a hold-out validation dataset approach, but some of the ideas can be extended to the multiclass problem as well as to k -fold cross-validation.

Some of these tools are in particular useful for *imbalanced* and/or *asymmetric* problems, which we will discuss later in this section. Remember that in binary classification, we have either $y = \{-1, 1\}$. If a binary classifier is used to detect the presence of something, such as a disease, an object on the radar, etc., the convention is that $y = 1$ (positive) denotes presence, and $y = -1$ (negative) denotes absence. This convention is the background for a lot of the terminology we will introduce now.

The Confusion Matrix and the ROC Curve

If we learn a binary classifier and evaluate it on a hold-out validation dataset, a simple yet useful way to inspect the performance beyond just computing $E_{\text{hold-out}}$ is a *confusion matrix*. By separating the validation data in four groups depending on y (the actual output) and $\hat{y}(\mathbf{x})$ (the output predicted by the classifier), we can make the confusion matrix:

	$y = -1$	$y = 1$	<i>total</i>
$\hat{y}(\mathbf{x}) = -1$	True neg (TN)	False neg (FN)	N^*
$\hat{y}(\mathbf{x}) = 1$	False pos (FP)	True pos (TP)	P^*
<i>total</i>	N	P	n

Of course, TN, FN, FP, and TP (and also N^* , P^* , N , P and n) should be replaced by the actual numbers, as in Example 4.5. Note that P (N) denotes the total number of positive (negative) examples in the data set, whereas P^* (N^*) denotes the total number of positive (negative) predictions made by the model. The confusion matrix provides a quick and informative overview of the characteristics of a classifier. For asymmetric problems, which we will soon introduce, it is important to distinguish between false positive (FP, also called *type I error*) and false negative (FN, also called *type II error*). Ideally they should both be 0, but in practice, there is usually a tradeoff between these two errors, and the confusion matrix is a helpful tool in visualising them both. That tradeoff between false negatives and false positives can often be done by tuning a decision threshold r , which is present in many binary classifiers (3.36).

There is also a wide body of terminology related to the confusion matrix, which is summarised in Table 4.1. Some particularly common terms are the

$$\text{recall} = \frac{TP}{P} = \frac{TP}{TP + FN} \quad \text{and the} \quad \text{precision} = \frac{TP}{P^*} = \frac{TP}{TP + FP}.$$

Table 4.1: Some common terms related to the quantities (TN, FN, FP, TP) in the confusion matrix. The terms written in italics are discussed in the text.

Ratio	Name
FP/N	False positive rate, Fall-out, Probability of false alarm
TN/N	True negative rate, Specificity, Selectivity
TP/P	True positive rate, Sensitivity, Power, <i>Recall</i> , Probability of detection
FN/P	False negative rate, Miss rate
TP/P*	Positive predictive value, <i>Precision</i>
FP/P*	False discovery rate
TN/N*	Negative predictive value
FN/N*	False omission rate
P/n	Prevalence
(FN + FP)/n	<i>Misclassification rate</i>
(TN + TP)/n	Accuracy, 1 – misclassification rate
$2TP/(P^* + P)$	F_1 score
$(1 + \beta^2)TP/((1 + \beta^2)TP + \beta^2FN + FP)$	F_β score

Recall describes how large a proportion of the positive data points are correctly predicted as positive. A high recall (close to 1) is good, and a low recall (close to 0) indicates a problem with many false negatives. Precision describes what the ratio of true positive points are among the ones predicted as positive. A high precision (close to 1) is good, and a low precision (close to 0) indicates a problem with many false positives.

Many classifiers contains a threshold r (3.36). If we want to compare different classifiers for a certain problem without specifying a certain decision threshold r , the *ROC curve* can be useful. The abbreviation ROC means ‘receiver operating characteristics’ and is due to its history from communications theory.

To plot an ROC curve, the recall/true positive rate (TP/P; a large value is good) is drawn against the false positive rate (FP/N; a small value is good) for all values of $r \in [0, 1]$. The curve typically looks as shown in Figure 4.13a. An ROC curve for a perfect classifier (always predicting the correct value for all $r \in (0, 1)$) touches the upper left corner, whereas a classifier which only assigns random guesses⁷ gives a straight diagonal line.

A compact summary of the ROC curve is the *area under the ROC curve*, *ROC-AUC*. From Figure 4.13a, we conclude that a perfect classifier has ROC-AUC = 1, whereas a classifier which only assigns random guesses has ROC-AUC = 0.5. The ROC-AUC thus summarises the performance of a classifier for *all* possible values of the decision threshold r in a single number.

The F_1 Score and the Precision–Recall Curve

Many binary classification problems have particular characteristics, in that they are imbalanced, or asymmetric, or both. We say that a problem is

⁷That is, predicts $\hat{y} = -1$ with probability r .

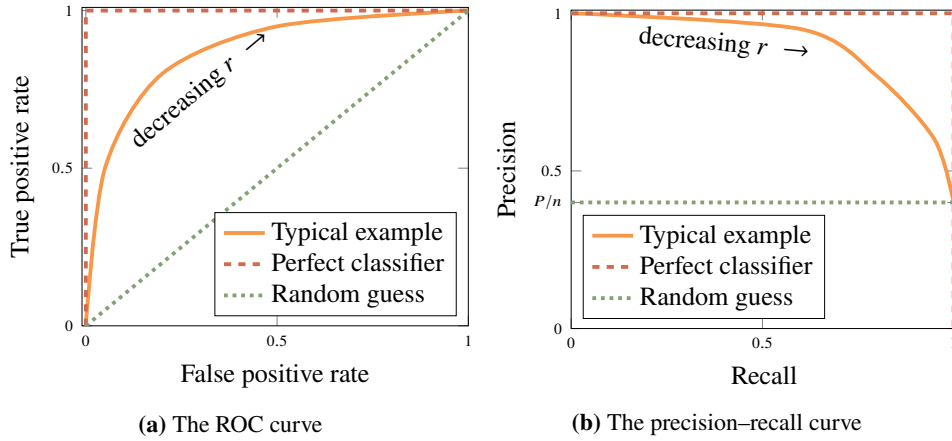


Figure 4.13: The ROC (left) and the precision-recall (right) curves. Both plots summarise the performance of a classifier for *all* decision thresholds r (see (3.36)), but the ROC curve is most relevant for balanced problems, whereas the precision-recall curve is more informative for imbalanced problems.

- (i) *imbalanced* if the vast majority of the data points belong to one class, typically the negative class $y = -1$. This imbalance implies that a (useless) classifier which always predicts $\hat{y}(\mathbf{x}) = -1$ will score very well in terms of misclassification rate (4.1a).
- (ii) *asymmetric* if a false negative is considered more severe than a false positive, or vice versa. That asymmetry is not taken into account in the misclassification rate (4.1a);

A typical imbalanced problem is the prediction of a rare disease (most patients do not have it – that is, most data points are negative). That problem could also be an asymmetric problem if it is, say, considered more problematic to predict an infected patient as healthy than vice versa.

To start with, the confusion matrix offers a good opportunity to inspect the false negatives and positives in a more explicit fashion. It can, however, sometimes be useful to also summarise the performance into a single number. For that purpose, the misclassification rate is not very helpful; in a severely imbalanced problem, it can, for instance, favour a useless predictor that always predicts -1 over any realistically useful predictor.

For imbalanced problems, where the negative class $y = -1$ is the most common class, the F_1 score is therefore preferable to the misclassification rate (or accuracy). The F_1 score summarises the precision and recall by their harmonic means,

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}, \quad (4.24)$$

which is a number between zero and one (higher is better).

For asymmetric problems, however, the F_1 score is not sufficient since it does not factor in the preference of having one type of error considered more serious than the other. For that purpose, a generalisation of the F_1 score, namely the F_β score, can be used. The F_β score weighs together precision and recall by considering recall to be β times as important as precision:

$$F_\beta = \frac{(1 + \beta^2) \text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}. \quad (4.25)$$

Much as the the misclassification rate might be misleading for imbalanced problems, the ROC curve might also be misleading for such problems. Instead, the precision–recall curve can (for imbalanced problems where $y = -1$ is the most common class) be more useful. As the name suggests, the precision–recall curve plots the precision (TP/P*; a large value is good) against the recall (TP/P; a large value is good) for all values of $r \in [0, 1]$, much like the ROC curve. The precision–recall curve for the perfect classifier touches the upper right corner, and a classifier which only assigns random guesses gives a horizontal line at the level P/n , as shown in Figure 4.13b.

Also for the precision–recall curve, we can define the *area under the precision–recall curve*, *PR-AUC*. The best possible PR-AUC is 1, and the classifier which only makes random guesses has PR-AUC equal to P/n .

We summarise this section with an example of an imbalanced and asymmetric problem in medicine. The evaluation of real-world classification problems, which most often are both imbalanced and asymmetric, is however, a challenging topic with certain dilemmas that are discussed further in Chapter 12.

Example 4.5 The confusion matrix in thyroid disease detection

The thyroid is an endocrine gland in the human body. The hormones it produces influences the metabolic rate and protein synthesis, and thyroid disorders may have serious implications. We consider the problem of detecting thyroid diseases, using the dataset provided by the UCI Machine Learning Repository (Dheeru and Karra Taniskidou 2017). The dataset contains 7 200 data points, each with 21 medical indicators as inputs (both qualitative and quantitative). It also contains the qualitative diagnosis {normal, hyperthyroid, hypothyroid}. For simplicity, we convert this into the binary problem with the output classes {normal, abnormal}. The dataset is split into training and hold-out validation sets, with 3 772 and 3 428 data points, respectively. The problem is imbalanced since only 7% of the data points are abnormal. Hence, the naive (and useless) classifier which always predicts normal will obtain a misclassification rate of around 7%. The problem is possibly also asymmetric, if false negatives (not indicating the disease) are considered more problematic than false positives (falsely indicating the disease). We train a logistic regression classifier and evaluate it on the validation dataset (using the default decision threshold $r = 0.5$, see (3.36)). We obtain the confusion matrix

	$y = \text{normal}$	$y = \text{abnormal}$
$\hat{y}(\mathbf{x}) = \text{normal}$	3 177	237
$\hat{y}(\mathbf{x}) = \text{abnormal}$	1	13

Most validation data points are correctly predicted as **normal**, but a large part of the **abnormal** data is also falsely predicted as normal. This might indeed be undesired in the application. The accuracy (1-misclassification) rate is 0.931, and the F_1 score is 0.106. (The useless predictor of always predicting **normal** has a very similar accuracy of 0.927, but worse F_1 score of 0.)

To change the picture, we lower the decision threshold to $r = 0.15$ in (3.36). That is, we predict the positive (**abnormal**) class whenever the predicted class probability exceeds this value, $g(\mathbf{x}) > 0.15$. This results in new predictions with the following confusion matrix:

	$y = \text{normal}$	$y = \text{abnormal}$
$\hat{y} = \text{normal}$	3 067	165
$\hat{y} = \text{abnormal}$	111	85

This change gives more true positives (85 instead of 13 patients are correctly predicted as **abnormal**), but this happens at the expense of more false positives (111 instead of 1 patients are now falsely predicted as **abnormal**). As expected, the accuracy is now lower at 0.919, but the F_1 score is higher at 0.381. Remember, however, that the F_1 score does not take the asymmetry into account but only the imbalance. We have to decide ourselves whether this classifier is a good tradeoff between the false negative and false positive rates, by considering which type of error has the more severe consequences.

4.6 Further Reading

This chapter was to a large extent inspired by the introductory machine learning textbook by Abu-Mostafa et al. (2012). There are also several other textbooks on machine learning, including Vapnik (2000) and Mohri et al. (2018), in which the central theme is understanding the generalisation gap using formal definitions of model flexibility such as the VC dimension or Rademacher complexity. The understanding of model flexibility for deep neural networks (Chapter 6) is, however, still subject to research; see for example, C. Zhang et al. (2017), Neyshabur et al. (2017), Belkin et al. (2019), and B. Neal et al. (2019) for some directions. Furthermore, the bias–variance decomposition is most often (including in this chapter) presented only for regression, but a possible generalisation to the classification problem is suggested by Domingos (2000). An alternative to the precision–recall curve, the so-called precision–recall–gain curve, is presented by Flach and Kull (2015).